

RECA MODEL



Structural Representation of Algorithmic
Complexity through Layered Architectures
and Selection Operators

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ABSTRACT

This work introduces the **RECA** model and places the researcher within an innovative formal framework that proposes a structural reinterpretation of computational complexity. Unlike classical theory based on the time evolution of the Turing machine, **RECA** shifts the analysis to the topology of the state space, modeling the computational process as a hierarchical organization of layers. The model defines computation using the tuple $R=(S, C, \Phi, \Xi)$, where the operator Ξ is introduced. This operator is structural and seeks symmetry properties that allow for bulk filtering (like interference in physics), not element-by-element filtering. It is a verification mechanism on the power set of state layers. Under this architecture, the class P is described as a unique sequential path, while the class NP is characterized by exponential growth of the layer space with an inherent symmetry structure. The central hypothesis of the model posits that solving complex problems lies in eliminating structural noise and identifying symmetry breaks within the search space, allowing the solution to emerge as a geometric relationship rather than the result of exhaustive search and location. This approach offers a new heuristic for developing efficient, "discrete" algorithms, with potential applications in Structural Artificial intelligence (AI), Cryptanalysis, and Bioinformatics.

Keywords: *Structural Complexity, State Space, P vs NP, Selection Operator, Algorithmic Symmetry, RECA Model.*

SUMMARY

The **RECA** (Structural Representation of Algorithmic Complexity) model proposes a formal framework for describing computational processes using a layered state architecture. Unlike the classical formulation based on the time evolution of the Turing Machine, **RECA** focuses on the structure of the computational state space, understood as a geometric organization of possible trajectories. The model distinguishes between deterministic processes, characterized by unique sequential trajectories, and non-deterministic processes, characterized by branching structures of state-space exploration. The Selection Operator (Ξ) is introduced as an abstract mechanism for structural verification on this space. This work proposes a structural interpretation of the complexity classes P and NP within a layered computation framework, emphasizing the organization of the search space over the purely temporal notion of computation.

1. Introduction

Classical computational complexity theory classifies problems based on the time resources required by deterministic and non deterministic Turing machines. However, this approach does not explicitly describe the internal structure of the state space underlying the computational process. **RECA** proposes an alternative representation based on the layered organization of the state space, where complexity is interpreted as a structural property of accessibility within that space.

2. Layered Computing System (LCS)

The Structural Distance Metric is defined as the distance (d) between two layers C_i and C_j as the measure of the uncertainty or "informational breadth" of the space of states between them. Formally: $d(C_i, C_j) = \log_2 |K|$ Where K is the set of possible paths that connect the states of layer C_i with those of layer C_j . In the P-Layer Architecture, there is a unique path between layers ($|K|=1$), therefore $d = 0$. This indicates that the complexity is purely sequential and There is no structural uncertainty.

In NP Layer Architecture, the number of paths grows exponentially ($|K|=2^{p(n)}$), which generates a structural distance $d = p(n)$.

This metric allows us to quantify "structural noise" that the Selection Operator (Ξ) must process. The objective of an efficient resolution under the Model **RECA** consists of applying a transformation that collapse this distance towards 0, transforming the exploration of a volume of states in a Linear trajectory. Therefore, in a **RECA** system, defines as a Tuple:

$$R = (S, C, \Phi, \Xi)$$

where:

S is the set of computational states
 $C = \{C_1, C_2, \dots, C_n\}$ is the set of existing layers:
 $\Phi: S \times C \rightarrow S \times C$ is the transition function between existing states and layers
 $\Xi: P(C) \rightarrow P(C)$ is a selection operator on the existing set of layers

3. P-Layer Architecture (Deterministic Computation)

If a layer is a state of memory, the model is clear. If a layer is a "possibility," then it's needed. Define whether that layer coexists (as in quantum computing) or if it is just a mathematical abstraction of branching. Then a system described by Layer P is characterized by a unique and sequential computational path:

$$S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \dots \rightarrow S_T$$

Polynomial Time Bound:

$$T(n) = O(n^k), \text{ con } k \in \mathbb{N}$$

Transition uniqueness: each state has at most one predecessor

Representation of complexity:

$$TP^{(n)} = a_0 + a_1 \cdot n + a_2 \cdot n^2 + \dots + a_k \cdot n^k$$

4. NP-Layer Architecture (Non-Deterministic Computation)

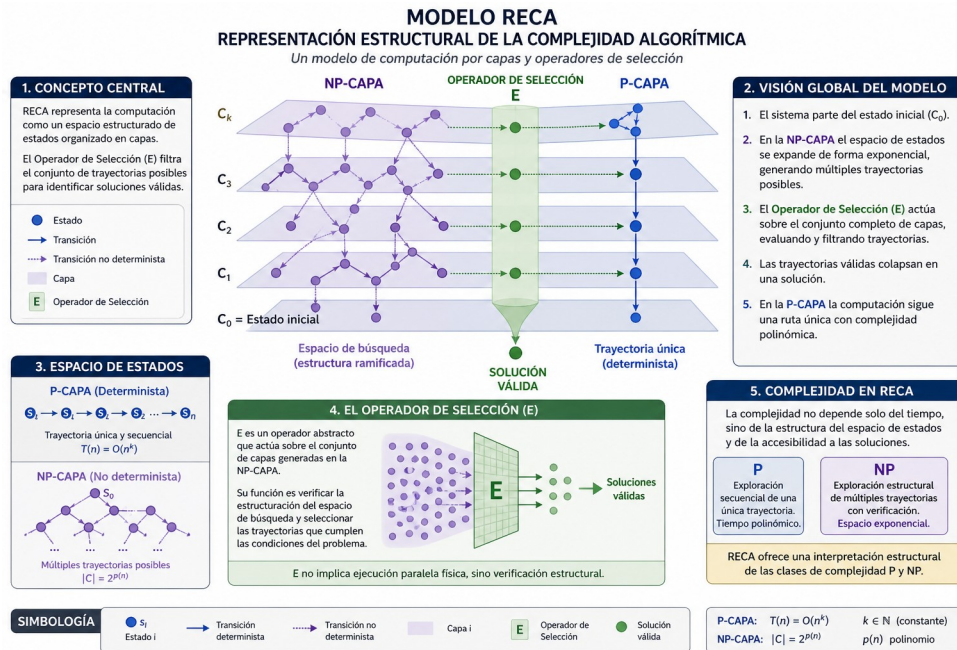


Figure 1. Structural representation of the RECA Model (Original Spanish version).

An NP-layer system is characterized by exponential growth of the layer space:

$$|C| = 2^{p(n)}, \text{ where } p(n) \text{ is a polynomial}$$

Each layer represents a possible path. computational within the search space

The selection operator Ξ acts on the entire set of layers to identify valid subsets

Guide to terms: NP-CAPA (NP-Layer); P-CAPA (P-Layer); Operador de Selección (Selection Operator Ξ); Espacio de búsqueda (Search Space); Trayectoria única (Unique trajectory); Solución válida (Valid solution); Estado inicial (Initial state); Ruido estructural (Structural noise).

Interpretation of the NP model

The NP Layer model does not imply physical parallel execution.

The layers represent an abstract structure of the computational search space.

The computational cost is not associated with the number of layers itself, but with the evaluation of the selection operator E on that structure.

5. Selection Operator Ξ

The operator Ξ is an abstract function that:

1. Filter layers of the state space
2. Select valid solutions
3. It represents a structural verification process

This operator is not interpreted as an instantaneous physical process, but as a mathematical transformation on the state space.

6. Interpretation of structural complexity

RECA proposes that computational complexity depends not only on execution time, but also on:

1. The internal structure of the state space
2. The organization of computational trajectories
3. The accessibility of solutions within the search space

7. Relationship with classical theory

RECA does not redefine the complexity classes P and NP , but offers a structural interpretation of them:

P corresponds to the sequential exploration of a single computational path

NP corresponds to the structured exploration of multiple paths with verification

8. Limitations of the model

This work does not intend to prove that $P \neq NP$, nor to formally redefine complexity classes, nor to replace the Turing Machine model. It is a representational and structural framework.

9. Conclusion

RECA introduces a structural perspective of computation based on layers of states and selection operators. This approach allows algorithmic complexity to be reinterpreted as a property of the state space, rather than solely as a time measure.

As a perspective for future development, the model **RECA** suggests that the core of the P vs NP problem could lie in the definition of a metric of symmetric collapse. If the class P is a unique path and NP is a layer space with inherent symmetry, resolving the complexity hierarchy would depend on determining whether there exists a mathematical transformation capable of collapsing the symmetry of NP into the sequence of P , without requiring exhaustive evaluation of each layer of the state space.

References

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